

Assessment Tool 3 - Descriptive Written Test

Question 1

A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. Explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Answer:

A Smart Home Automation System uses Artificial Intelligence (AI) to automatically manage lighting, temperature and security using sensor information.

Simple Reflex Agent: Acts only on current inputs. Example: turns lights ON when motion is detected and activates alarms when an intruder is detected. It is simple and fast but cannot learn from past events.

Model-Based Agent: Maintains an internal model of the environment. It remembers room occupancy, previous temperature settings and door status to make better decisions.

Goal-Based Agent: Makes decisions to achieve goals such as maintaining comfortable temperature, reducing electricity consumption and improving security.

Utility-Based Agent: Chooses the action that provides the highest overall benefit by balancing comfort, energy efficiency and security. It can also adapt to user preferences over time.

Justification: A hybrid approach combining Model-Based and Utility-Based agents is the best choice because it remembers previous information, learns user preferences and selects the most beneficial action for comfort, energy efficiency and security.

Question 2

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Answer:

Uniform Cost Search (UCS) expands the path with the lowest cumulative cost first and always finds the optimal path when costs are positive. Iterative Deepening Search (IDS) repeatedly performs depth-limited searches until the goal is found, combining the advantages of Depth First Search and Breadth First Search.

Advantages of UCS: Finds the least-cost path and guarantees an optimal solution.

Disadvantages of UCS: Requires more memory and can be slower when many paths exist.

Advantages of IDS: Uses very little memory, is complete for finite search spaces and is suitable when the search depth is unknown.

Disadvantages of IDS: Repeats searches at each depth level, increasing computation time.

A Goal-Based or Model-Based intelligent agent is suitable because it evaluates the environment before selecting actions. Combining a Goal-Based Agent with Uniform Cost Search is the most effective solution because the agent focuses on reaching the goal while UCS guarantees the least-cost path to the exit.